7th Grade Unit 8 Assessment Guide

General Information	This assessment covers three benchmarks: MA.7.NSO.1.1, MA.7.NSO.2.1, and MA.7.NSO.2.2. These three benchmarks are interconnected. MA.7.NSO.2.2 addresses fluency of operations with rational numbers. This fluency was a focus of Unit 3. The expectations of procedural fluency for grade 7 students builds on the fluency expectations of grade 6. "In grade 6, students performed operations with integers, multiplied and divided positive multi-digit numbers with decimals to the thousandths and computed products and quotients of positive fractions by positive fractions, including mixed numbers with procedural fluency." *-from the Grade 7 B.E.S.T. Instructional Guide for Mathematics* Because this is the expectation of grade 6 students, the procedural fluency expectations by the end of grade 7 build on this so that students in grade 7 should be able to perform all four operations with rational numbers, including negative values, with procedural fluency. Teachers are encouraged to revisit the unit assessment guide for Unit 3 to review the information provided on fluency. The benchmark MA.7.NSO.1.1 is very close to MA.6.NSO.3.3. In grade 7, students learn zero exponent law and explore the other laws of exponents with whole number exponents, but there is an expectation of some overlap with grade 6. One difference is that MA.7.NSO.1.1 allows for positive and negative rational numbers whereas MA.6.NSO.3.3 only allows for positive rational numbers. Some of the questions on this assessment could be considered aligned to both benchmarks. When teachers consider students who struggle with the Laws, noticing whether students are struggling with the introduction of negative rational numbers as bases could be helpful to determine the type of support a student may need. If a student can perform the law with a positive base but not with a negative base, then the student will need support on operations with integers (see Grade 6, Unit 7 or Grade 7, Unit 1, Lesson 4). The guide includes the names of the laws of exponents that each question assesses, where applicable. Teachers may want to use this feature to note which laws students are struggling on so that support can be more focused. Students apply their knowledge of the Laws of Exponents within the benchmark MA.7.NSO.2.1. Teachers are encouraged to evaluate student work to determine whether student mistakes are misunderstanding what operations should be applied first or if the students do not understand the Laws of Exponents or absolute value. If students need support on absolute value, consider using Grade 6, Unit 4, Lesson 12. To support students who struggle with operations with rational numbers consider using Grade 6, Unit 2, Lesson 9 or Grade 7, Unit 3.
Calculator Information	The questions on this assessment are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	**1**	**Benchmark(s)**	MA.7.NSO.1.1; MA.7.NSO.2.2
729		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
		Law(s) of Exponents	• Power of a power
Question	**2**	**Benchmark(s)**	MA.7.NSO.1.1; MA.7.NSO.2.2
50.41		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**3**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
-3.57		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**4**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
56.5		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

<table>
<tr><td>Question</td><td>5</td><td>Benchmark(s)</td><td>MA.7.NSO.1.1; MA.7.NSO.2.2</td></tr>
<tr><td colspan="2" rowspan="2">$-\frac{1}{128}$</td><td>Recommended Scoring

(5 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Law(s) of Exponents</td><td>• Product of powers</td></tr>
<tr><td>Question</td><td>6</td><td>Benchmark(s)</td><td>MA.7.NSO.2.1; MA.7.NSO.2.2</td></tr>
<tr><td colspan="2" rowspan="2">125</td><td>Recommended Scoring

(5 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Law(s) of Exponents</td><td>• Zero exponent</td></tr>
<tr><td>Question</td><td>7</td><td>Benchmark(s)</td><td>MA.7.NSO.1.1; MA.7.NSO.2.2</td></tr>
<tr><td colspan="2" rowspan="2">-15.625</td><td>Recommended Scoring

(5 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Law(s) of Exponents</td><td>• Quotient of powers</td></tr>
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Question	8	Benchmark(s)	MA.7.NSO.2.1; MA.7.NSO.2.2
230.4		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. • Identity exponent
Question	**9**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
12.04		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**10**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
$\frac{38}{7}$		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**11**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
$-\frac{8}{27}$		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	12	Benchmark(s)	MA.7.NSO.1.1; MA.7.NSO.2.2
-1		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**13**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
☐ $\frac{-18}{10}$ ☐ $-\frac{9}{5}$ ☐ $\frac{216}{32}$ ☐ $\frac{{}^{-}6\cdot{}^{-}6\cdot{}^{-}6}{2\cdot2\cdot2\cdot2\cdot2}$ ☐ $-3\cdot-3\cdot-3\cdot\frac{1}{2}\cdot\frac{1}{2}$		**Recommended Scoring** **(10 Total Points)**	Consider 5 points for each correct answer. Consider deducting 3 points for each incorrect answer. Consider no points if all are selected.
Question	**14**	**Benchmark(s)**	MA.7.NSO.2.1; MA.7.NSO.2.2
☐ $8\cdot(-6)^6$ ☐ $8\cdot3(64)$ ☐ $24\cdot(-2)^6$ ☐ $8\cdot3(-12)$ ☐ $2^3\cdot(-2)^6\cdot3$		**Recommended Scoring** **(15 Total Points)**	Consider 5 points for each correct answer. Consider deducting 3 points for each incorrect answer. Consider no points if all are selected.

Question	15	Benchmark(s)	MA.7.NSO.2.1; MA.7.NSO.2.2
☐ -11 ☐ 0 ☐ $\frac{64}{-8}+\frac{6}{-2}$ ☐ $\frac{64}{16}+\frac{8}{-2}$ ☐ $8^2 \div \left[(-4)^2+\left(-\frac{1}{2}\right)\right](2^3)$ ☐ $\left[8^2 \div (-4)^2+\left(-\frac{1}{2}\right)\right](2^3)$ ☐ $(8^2 \div (-4)^2)+\left(\left(-\frac{1}{2}\right)(2^3)\right)$		**Recommended Scoring** **(15 Total Points)**	Consider 5 points for each correct answer. Consider deducting 3 points for each incorrect answer. Consider no points if all are selected.